

AMAZON SALES DATA ANALYSIS

Task 4 – Data Storytelling & Statistical Validation Report

Prepared By: G. Bhanupriya

Internship Program: Data Analytics Internship

Organization: ApexPlanet Software Pvt. Ltd.

Project Title: Amazon Sales Data Analysis

Submission Date: June 2026

EXECUTIVE SUMMARY

The Amazon Sales Data Analysis project was carried out as part of the Data Analytics Internship Program to transform raw sales data into meaningful business insights through data storytelling and statistical validation. The project demonstrates how data analytics techniques can be used to understand customer purchasing behavior, product performance, sales trends, regional performance, and payment preferences. The analysis was performed using Python, SQL, Power BI, and statistical methods to convert raw business data into actionable insights that support strategic business decisions.

The project includes data preprocessing, exploratory data analysis (EDA), KPI development, dashboard visualization, and statistical validation. The findings were then presented as a business story to effectively communicate analytical results to both technical and non-technical stakeholders. The overall analysis demonstrates how data-driven decision-making can improve business performance and customer satisfaction.

1. PROJECT OBJECTIVE

The primary objective of this project is to transform Amazon sales data into meaningful business insights through data storytelling while validating important analytical findings using statistical techniques.

Specific Objectives

- Analyze overall sales performance.
- Identify high-performing product categories.
- Study customer purchasing behavior.
- Evaluate regional sales performance.

- Analyze customer payment preferences.
- Develop an interactive business dashboard.
- Validate analytical findings using statistical testing.
- Provide business recommendations for improved decision-making.

2. BUSINESS PROBLEM

Amazon serves customers across multiple regions through numerous product categories. Understanding customer purchasing patterns, product performance, regional sales, and payment preferences is essential for improving profitability and operational efficiency.

This project attempts to answer the following business questions:

- Which product categories generate the highest sales revenue?
- Which products contribute the most to business growth?
- Which regions generate the highest sales?
- Which payment methods are preferred by customers?
- How do sales change over different months?
- What strategies can improve overall business performance?

3. DATA STORY

The analysis began by collecting and preparing the Amazon sales dataset. Data preprocessing techniques were applied to improve data quality by removing duplicate records, handling missing values, correcting data types, and standardizing column names. These preprocessing activities ensured reliable analytical results.

Exploratory Data Analysis (EDA) was then performed using Python, SQL, and visualization libraries to understand sales performance, customer purchasing behavior, payment preferences, and regional trends. Multiple visualizations including bar charts, line charts, pie charts, heatmaps, scatter plots, histograms, and boxplots were created to explore the data from different business perspectives.

An interactive Power BI dashboard was developed to summarize key business metrics using KPI cards and visualizations. Finally, statistical hypothesis testing was performed to validate one of the important business findings and strengthen the reliability of the analysis.

The project successfully transformed raw business data into meaningful business intelligence that supports strategic decision-making.

4. DATA ANALYSIS SUMMARY

- Several analytical techniques were applied throughout the project.
- Sales Performance Analysis
- Sales performance was analyzed across different product categories to identify high-performing business segments.
- Customer Behavior Analysis
- Customer purchasing patterns were examined using order frequency, payment preferences, and purchasing trends.
- Regional Sales Analysis
- Revenue generated across different locations was compared to identify regions contributing the highest sales.
- Payment Method Analysis
- Customer payment preferences were analyzed to understand the popularity of different payment methods.
- Monthly Sales Trend Analysis
- Monthly sales performance was evaluated to identify seasonal business trends and revenue fluctuations.

5. DASHBOARD SUMMARY

An interactive Power BI dashboard was developed to visualize important business metrics and provide decision-makers with a comprehensive overview of business performance.

The dashboard includes the following components:

- KPI Cards
- Total Sales
- Total Orders
- Average Order Value (AOV)
- Customer Count
- Category-wise Sales
- Product-wise Revenue
- Monthly Sales Trend
- Regional Sales Distribution
- Payment Method Analysis
- Customer Rating Analysis

The dashboard provides an interactive and user-friendly representation of the analytical findings.

6. KEY FINDINGS

The analysis generated several valuable business insights.

- Electronics emerged as the highest revenue-generating product category.
- A limited number of products contributed a significant portion of the total revenue.
- Monthly sales exhibited seasonal variations with higher revenue during promotional periods.
- UPI and Credit Card were among the most preferred payment methods.
- Certain regions consistently generated higher sales than others.
- Products receiving higher customer ratings generally achieved stronger sales performance.
- Customer purchasing behavior varied across different product categories and regions.

7. STATISTICAL VALIDATION

To strengthen the reliability of the analysis, statistical hypothesis testing was performed.

Business Question

Is there a significant difference in average sales between Electronics and Clothing product categories?

Null Hypothesis (H_0)

There is no statistically significant difference in average sales between Electronics and Clothing product categories.

Alternative Hypothesis (H_1)

There is a statistically significant difference in average sales between Electronics and Clothing product categories.

Statistical Test Used

Independent Samples T-Test

Decision Rule

- If the p-value is less than 0.05, reject the Null Hypothesis.
- If the p-value is greater than or equal to 0.05, fail to reject the Null Hypothesis.

Interpretation

The statistical test determines whether the observed difference in average sales between the selected product categories is statistically significant or occurred by chance. Statistical validation increases confidence in the analytical findings and supports evidence-based business decision-making.

8. BUSINESS INSIGHTS

The analysis provides several meaningful business insights.

- High-performing product categories should receive greater inventory allocation.
- Marketing campaigns should focus on regions with lower sales performance.
- Digital payment methods should continue to be promoted to improve customer convenience.
- Seasonal sales trends should be considered while planning inventory and promotional activities.
- Highly rated products should receive increased promotional visibility.
- Low-performing products should be monitored and evaluated for improvement opportunities.

9. BUSINESS RECOMMENDATIONS

Based on the findings obtained from the analysis, the following recommendations are proposed.

- Increase inventory levels for best-selling products.
- Strengthen marketing activities in low-performing regions.
- Introduce promotional offers during seasonal demand periods.
- Improve customer service for products receiving lower ratings.
- Encourage customer loyalty through reward programs.
- Continue leveraging data analytics to monitor sales performance and customer behavior.
- Regularly update dashboards to support real-time business decision-making.

10. BUSINESS IMPACT

The recommendations developed through this project can significantly improve business performance by:

- Increasing revenue generation.
- Improving inventory management and operational efficiency.
- Enhancing customer satisfaction.
- Optimizing marketing strategies.
- Supporting better business planning.
- Encouraging data-driven decision-making.

11. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used throughout the project:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SQL (SQLite)
- SciPy
- Power BI
- VS Code
- GitHub
- Microsoft PowerPoint

12. CONCLUSION

The Amazon Sales Data Analysis project successfully demonstrates how modern data analytics techniques can transform raw business data into valuable business insights. Through data preprocessing, exploratory data analysis, dashboard visualization, KPI development, and statistical validation, the project provides a comprehensive understanding of customer purchasing behavior, sales performance, product trends, and regional business growth.

The data storytelling approach effectively communicates analytical findings in a simple and meaningful manner, allowing stakeholders to understand business performance quickly. Statistical validation further strengthens the credibility of the analysis by confirming that important findings are supported by evidence rather than random variation.

Overall, this project demonstrates the practical application of Python, SQL, Power BI, and statistical analysis in solving real-world business problems. The insights and recommendations generated through this analysis can support strategic decision-making, improve operational efficiency, enhance customer satisfaction, and contribute to sustainable business growth. The project also reflects strong analytical, visualization, problem-solving, and communication skills that are essential for a successful career in Data Analytics.

13. REFERENCES

- Amazon Sales Dataset
- Python Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Seaborn Documentation
- SciPy Documentation
- SQL Documentation
- Power BI Documentation
- GitHub Documentation